

Yusuf Abdi

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Professional Summary — Machine Learning Researcher with 3+ years of experience developing deep learning and computer vision solutions for healthcare and medical imaging applications. Experienced in the full machine learning lifecycle, including data preparation, annotation, preprocessing, model development, training, validation, deployment support, and performance evaluation. Proven track record collaborating with clinicians, researchers, and cross-functional engineering teams to develop AI-driven solutions for medical image analysis. Strong background in Python, deep learning frameworks, computer vision, medical imaging, and applied research, with published work in automated image segmentation and diagnostic AI systems.

Skills

Technical: Machine Learning Deep Neural Networks, Image Registration, Feature Extraction, Model Training, Model Validation, Research AND Development, Scientific Computing and Framework, Computer Vision control, NLP, LLM, CNN, UNit, Docker/Podman , Data labeling, Image Segmentation, Semantic Augmentation, Data Handling, Data-mining and analysis, Object Detection, Classification Models, Transfer Learning, along with a solid understanding of data structures, algorithms, and Web Development.	Model Evaluation: Cross-Validation, ROC-AOC, Precision, Recall, F1-Score, Dice Similarity Coefficient, Intersection over Union(IoU), Error Analysis, Model Optimization.
Programming: Python, JavaScript, Java, C, C++, SQL, R,	Medical Imaging: CT Imaging, Medical Image Segmentation, DICOM Data Processing, Radiographic Image Analysis, 2D/3D Medical Imaging, Image Annotation, Data Augmentation, Clinical Dataset Management
	Frameworks/Tools: PyTorch, TensorFlow, Keras, OpenCV, Scikit-Learning, NumPy, Pandas, Git/Github, Docker, Jupyter Notebook, Databases

Experience

Adjunct Faculty, Computer Science
Bridgewater State University

Sep 2025 – Present

- COMP 151 Computer Science -I
- COMP 152 Computer Science -II
- COMP 350 Operating Systems
- COMP 430 Computer Networks

Adjunct Faculty, Computer Science
Framingham State University

Jan 2026 – Present

- CSC 465 Operating Systems Internals

Machine Learning Engineer, Research Collaborator
Mass General Brigham Women's Hospital, Radiology Department

Oct 2022 – Feb 2026

- Design and develop deep learning models for automation in medical image analysis with 98 percent Accuracy.
- Managed large-scale medical imaging datasets and supported model development using clinical imaging data.
- Prepare and process various large datasets of medical images to be compatible for Machine Learning; tasks included: labeling, normalization, augmentation, and annotation; deployed in the cloud, maintained GitHub repositories, and resolved compatibility issues. Collaborate with the radiology doctor to understand clinical requirements and translate them into an AI-driven solution. Contributed to the research outcome paper on the technical procedure.

Career Services Manager
International Institute of New England (IINE)

Apr 2021 – Present

- Manage the employment department and employer partnerships, leading to over 1000 successful job placements and increased clients' self-sufficiency.
- Supervise a team of 13 Employment Specialists, 3 interns, and 4 professional volunteers (including physicians, engineers, HR professionals, and professors).
- Designed and implemented strategic goals for the employment department to align with organizational and client outcomes. Oversee compliance with federal, state, and internal reporting requirements, ensuring timely submission of employment outcomes and enrollment data.

(Temp assignment) Machine Learning/Software Engineer <i>LotusAI-New-startup-Martin Trust Center for MIT Entrepreneurship</i>	May 2025 – Aug 2025
– Led development of AI voice stack for clinical trial platform, integrating LLMs (OpenAI, RAG), STT/TTS systems (Whisper, ElevenLabs), and Twilio for voice-first conversational agents; collaborated cross-functionally on product, UX, and healthcare compliance to rapidly prototype and deliver end-to-end AI pipelines.	
Teaching Assistant, Data Mining Techniques (Supervised/Unsupervised ML/NLP) <i>Khoury College Northeastern University</i>	Jan 2024 – May 2025
– Held office hours, created draft assignments, quizzes, and graded, and managed student discussion forums. – Taught lab sessions for the course with over 100 students. Served as lead teaching assistant, managed, assisted, and oversaw course logistics and instructional duties.	
Employment Specialist <i>International Institute of New England (IINE)</i>	Jan 2014 – Apr 2021
Employment Counselor <i>U.S. Committee for Refugees and Immigrants/ Vermont Refugee Resettlement Program, Colchester, VT</i>	Jul 2009 – Dec 2013
Interpreter, Somali-English and Arabic-English <i>U.S. Committee for Refugees and Immigrants/ Vermont Refugee Resettlement Program, Colchester, VT</i>	Jul 2009 – Dec 2013
Computer Instructor <i>Fletcher Free Library, Burlington, VT</i>	Apr 2009 – Dec 2013
– Prepared lesson plans and taught foundation computer literacy to students pursuing GED and HiSET.	

Education

Northeastern University <i>M.S of Computer Science</i> <i>Major: Artificial Intelligence, Machine Learning, Computer Vision, Software Engineer</i> <i>Minor: Computer Science</i>	2021-2025
University of Massachusetts Lowell <i>Master Program of Peace and Conflict Studies</i> <i>Major: Social Economic Development</i> <i>Minor: Community Conflict Resolution</i> <i>Thesis: Prospect of Labor Integration: Somali Mainers in Housekeeping, Caregiving, and Farming, Supervised: DR.Thomas Piñeros Shields, Ph.D., (Thesis Advisor) Assistant Teaching Professor, Department of Sociology.</i>	2015-2019
Middlesex Community College <i>Associate Math</i>	2019-2020
Mogadishu University <i>Bachelor of Economics and Science</i>	1999-2003

Research Projects

WikiNews: User-Generated News and Content Management System	Spring 2025
– WikiNews enables users to create, share, update, and manage collaborative online news.	
Understanding Behavioral and Physiological Dynamics Through Wearable Monitoring.	Fall 2024
– This study aims to leverage mobile sensor data, especially accelerometer signals, to detect human physical activity, behavior, and sleep.	
Clinical Assertion Model Natural Language Processing	Spring 2024
– Developed a supervised NLP-based Clinical Assertion Model to extract key information from clinical notes, improving assertion classification accuracy and supporting more accurate, timely decision-making in patient care.	
Assessment of Machine Learning Algorithms Used in Classification of Breast Cancer Tumors	Spring 2023
– Used Machine Learning models to classify breast cancer; Neural Network achieved highest accuracy at 98.2 percent, improving diagnostic precision and outcomes.	
Database Management Systems Built a MySQL database to analyze FAA bird strike data effectively.	Fall 2022
– Designed and implemented MySQL and SQLite databases to analyze bird strikes and mine transactional data using star schema for analytical queries.	

Optimizing Northeastern's RedEye Shuttle Routes Using Minimum Spanning Tree Algorithms: A Cost-Efficient and Time-Saving Approach **Summer 2022**

- Northeastern University is the original node Used Euclidean distance formula, calculated the weight. The approaches, Queue, TSP: Prim's Algorithm, Minimum-weight Spanning Tree, Kruskal's Algorithm, the goal was to find the lowest time complexity, which was Kruskal's algorithm $O(V^2)$

Network Traffic Analyze in Java programming language **Fall 2021**

- Network Traffic Analyzer is a package analyzer software that screens all network movement. The project is written in the Java programming language. The application admin of the system can take network packets and analyze data received/sent from/to the network. System performance will be enhanced, and traffic will be controlled.

Selected Publications

Automated Detection and Segmentation of Ascending Aorta Dilation on a Non-ECG-Gated Chest CT Using Deep Learning

Authors: Fargana Aghayeva, Yusuf Abdi, Ahmad Uzair, Ayaz Aghayev